

STPS20M60D Datasheet



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DiGi Electronics Part Number	STPS20M60D-DG
Manufacturer	STMicroelectronics
Manufacturer Product Number	STPS20M60D
Description	DIODE SCHOTTKY 60V 20A TO220AC
Detailed Description	Diode 60 V 20A Through Hole TO-220AC

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Purchase and inquiry

Manufacturer Product Number:

STPS20M60D

Series:

-

Technology:

Schottky

Current - Average Rectified (Io):

20A

Speed:

Fast Recovery $\leq 500\text{ns}$, $> 200\text{mA}$ (Io)

Capacitance @ Vr, F:

-

Package / Case:

TO-220-2

Operating Temperature - Junction:

150°C (Max)

Manufacturer:

STMicroelectronics

Product Status:

Active

Voltage - DC Reverse (Vr) (Max):

60 V

Voltage - Forward (Vf) (Max) @ If:

580 mV @ 20 A

Current - Reverse Leakage @ Vr:

125 μA @ 60 V

Mounting Type:

Through Hole

Supplier Device Package:

TO-220AC

Base Product Number:

STPS20

Environmental & Export classification

RoHS Status:

ROHS3 Compliant

REACH Status:

REACH Unaffected

HTSUS:

8541.10.0080

Moisture Sensitivity Level (MSL):

1 (Unlimited)

ECCN:

EAR99



STPS20M60

Power Schottky rectifier

Features

- High current capability
- Avalanche rated
- Low forward voltage drop
- High frequency operation

Description

The STPS20M60D is a single diode Schottky rectifier, suited for high frequency switch mode power supply.

Packaged in TO-220AC, this device is intended to be used in notebook, game station and desktop adapters, providing in these applications a good efficiency at both low and high load.

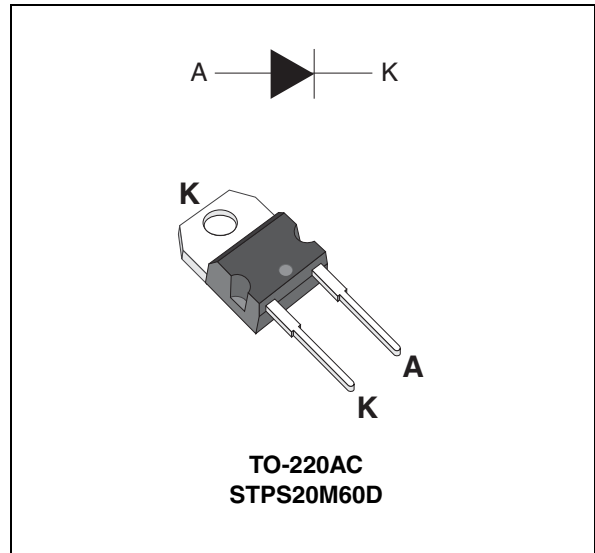
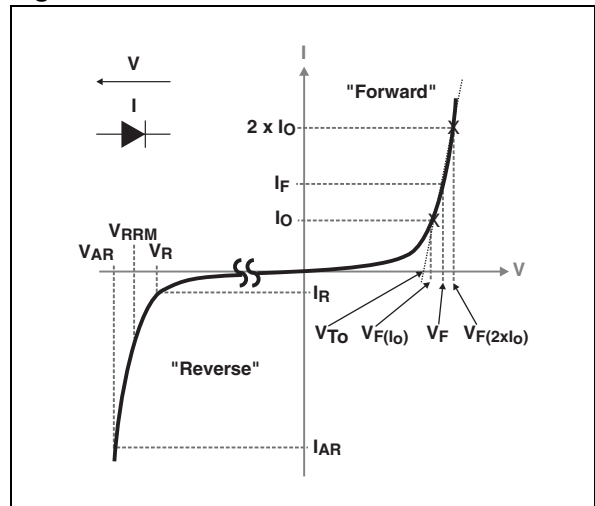


Table 1. Device summary

Symbol	Value
$I_{F(AV)}$	20 A
V_{RRM}	60 V
V_F (typ)	0.37 V
T_j (max)	150 °C

Figure 1. Electrical characteristics(a)



- a. V_{ARM} and I_{ARM} must respect the reverse safe operating area defined in [Figure 11](#). V_{AR} and I_{AR} are pulse measurements ($t_p < 1 \mu s$). V_R , I_R , V_{RRM} and V_F are static characteristics

1 Characteristics

Table 2. Absolute ratings (limiting values, at $T_{amb} = 25\text{ °C}$ unless otherwise specified)

Symbol	Parameter		Value	Unit
V_{RRM}	Repetitive peak reverse voltage		60	V
$I_{F(RMS)}$	Forward rms current		60	A
$I_{F(AV)}$	Average forward current, $\delta = 0.5$	$T_c = 135\text{ °C}$	20	A
I_{FSM}	Surge non repetitive forward current	$t_p = 10\text{ ms sine-wave}$	400	A
$P_{ARM}^{(1)}$	Repetitive peak avalanche power	$T_j = 25\text{ °C}, t_p = 1\text{ }\mu\text{s}$	26400	W
$V_{ARM}^{(2)}$	Maximum repetitive peak avalanche voltage	$t_p < 1\text{ }\mu\text{s}, T_j < 150\text{ °C}, I_{AR} < 99\text{ A}$	80	V
$V_{ASM}^{(2)}$	Maximum repetitive peak avalanche voltage	$t_p < 1\text{ }\mu\text{s}, T_j < 150\text{ °C}, I_{AR} < 99\text{ A}$	80	V
T_{stg}	Storage temperature range		-65 to +175	°C
T_j	Maximum operating junction temperature ⁽³⁾		150	°C

1. For temperature or pulse time duration deratings, please refer to [Figure 4](#) and [5](#). More details regarding the avalanche energy measurements and diode validation in the avalanche are provided in the application notes AN1768 and AN2025.

2. See [Figure 11](#)

3. $\frac{dP_{tot}}{dT_j} < \frac{1}{R_{th(j-a)}}$ condition to avoid thermal runaway for a diode on its own heatsink

Table 3. Thermal parameters

Symbol	Parameter	Value	Unit
$R_{th(j-c)}$	Junction to case	1.0	°C/W

STPS20M60

Characteristics

Table 4. Static electrical characteristics

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Unit
$I_R^{(1)}$	Reverse leakage current	$T_j = 25\text{ }^{\circ}\text{C}$	-	30	125	μA
		$T_j = 125\text{ }^{\circ}\text{C}$		20	75	mA
$V_F^{(2)}$	Forward voltage drop	$T_j = 25\text{ }^{\circ}\text{C}$	-	0.470	0.505	V
		$T_j = 125\text{ }^{\circ}\text{C}$	-	0.370	0.415	
		$T_j = 25\text{ }^{\circ}\text{C}$	-	0.530	0.580	
		$T_j = 125\text{ }^{\circ}\text{C}$	-	0.460	0.530	

1. Pulse test: $t_p = 5\text{ ms}$, $\delta < 2\%$ 2. Pulse test: $t_p = 380\text{ }\mu\text{s}$, $\delta < 2\%$

To evaluate the conduction losses use the following equation:

$$P = 0.385 \times I_{F(AV)} + 0.0073 \times I_{F(RMS)}^2$$

Figure 2. Average forward power dissipation versus average forward current

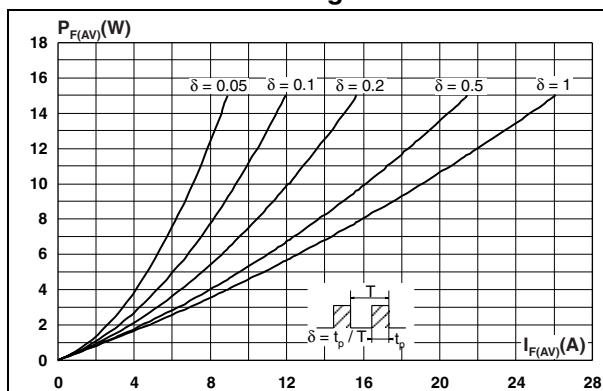
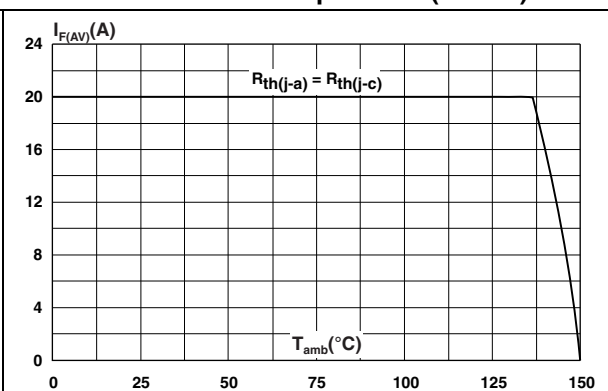
Figure 3. Average forward current versus ambient temperature ($\delta = 0.5$)

Figure 4. Normalized avalanche power derating versus pulse duration

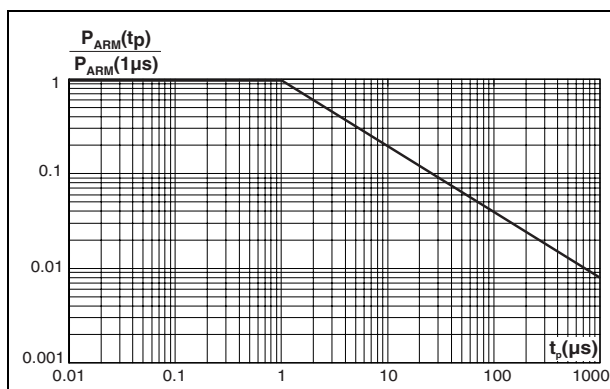
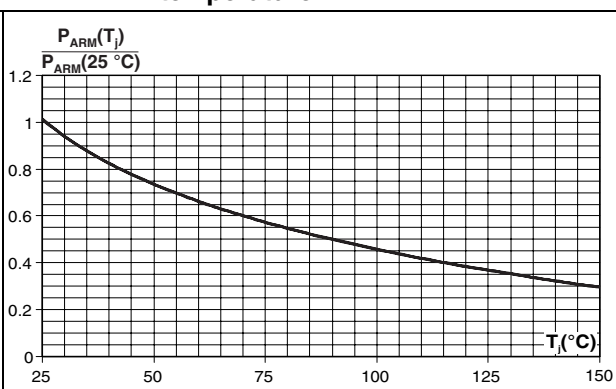


Figure 5. Normalized avalanche power derating versus junction temperature



Characteristics

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Figure 6. Non repetitive surge peak forward current versus overload duration (maximum values)

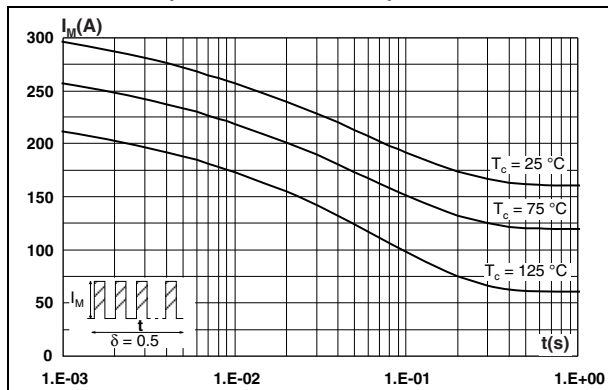


Figure 7. Relative thermal impedance junction to case versus pulse duration

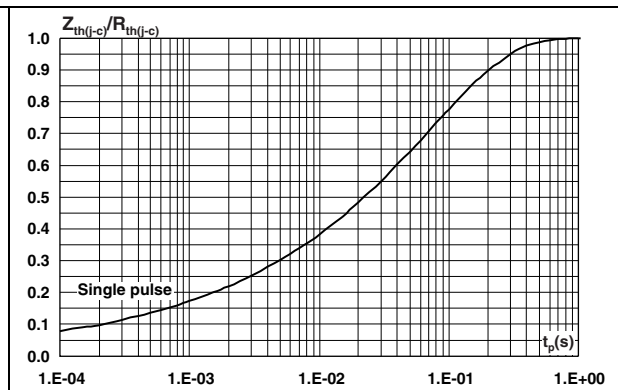


Figure 8. Reverse leakage current versus reverse voltage applied (typical values)

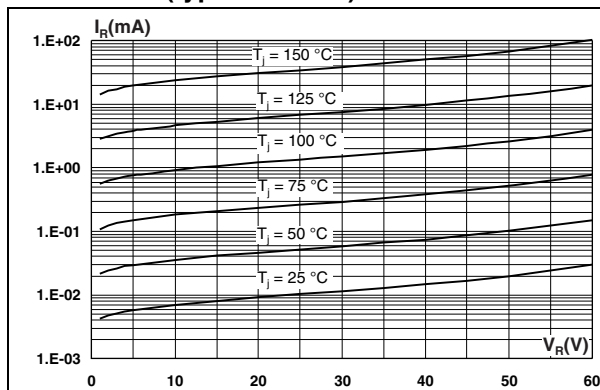


Figure 9. Junction capacitance versus reverse voltage applied (typical values)

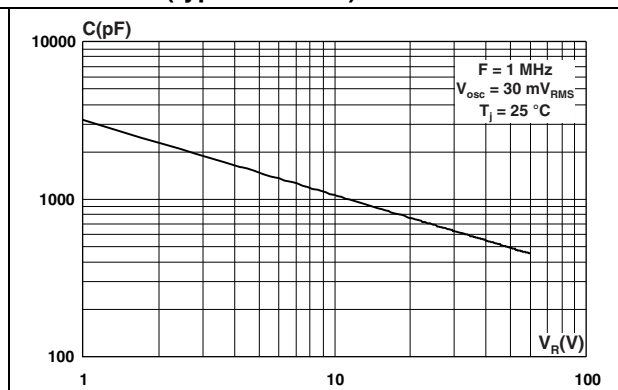


Figure 10. Forward voltage drop versus forward current

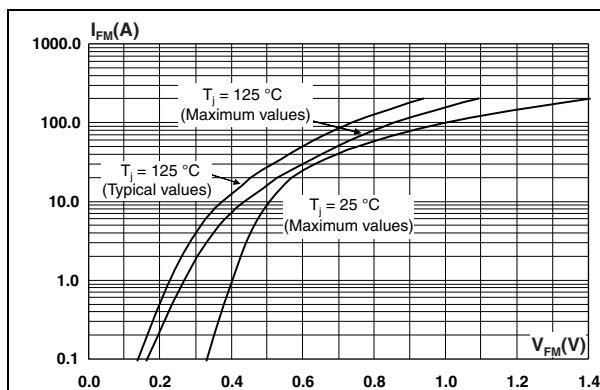
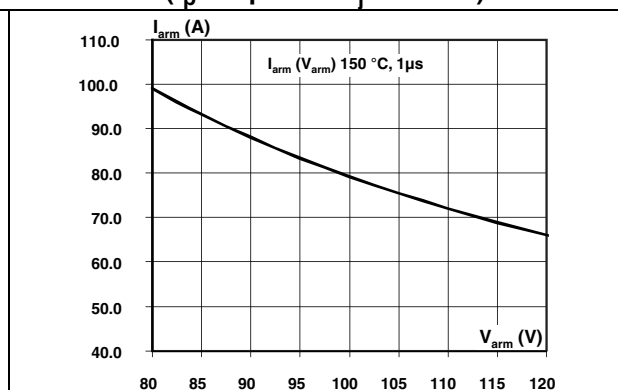


Figure 11. Reverse safe operating area ($t_p < 1 \mu s$ and $T_j < 150 \text{ °C}$)



2 Package information

- Epoxy meets UL94, V0
- Cooling method: by conduction (C)
- Recommended torque value: 0.4 to 0.6 N·m

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: www.st.com. ECOPACK® is an ST trademark.

Table 5. TO-220AC dimensions

Ref.	Dimensions			
	Millimeters		Inches	
	Min.	Max.	Min.	Max.
A	4.40	4.60	0.173	0.181
C	1.23	1.32	0.048	0.051
D	2.40	2.72	0.094	0.107
E	0.49	0.70	0.019	0.027
F	0.61	0.88	0.024	0.034
F1	1.14	1.70	0.044	0.066
G	4.95	5.15	0.194	0.202
H2	10.00	10.40	0.393	0.409
L2	16.40 typ.		0.645 typ.	
L4	13.00	14.00	0.511	0.551
L5	2.65	2.95	0.104	0.116
L6	15.25	15.75	0.600	0.620
L7	6.20	6.60	0.244	0.259
L9	3.50	3.93	0.137	0.154
M	2.6 typ.		0.102 typ.	
Diam. I	3.75	3.85	0.147	0.151

3 Ordering information

Table 6. Ordering information

Order code	Marking	Package	Weight	Base qty	Delivery mode
STPS20M60D	STPS20M60D	TO-220AC	1.86 g	50	Tube

4 Revision history

Table 7. Revision history

Date	Revision	Changes
02-Nov-2011	1	First issue.

STPS20M60

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